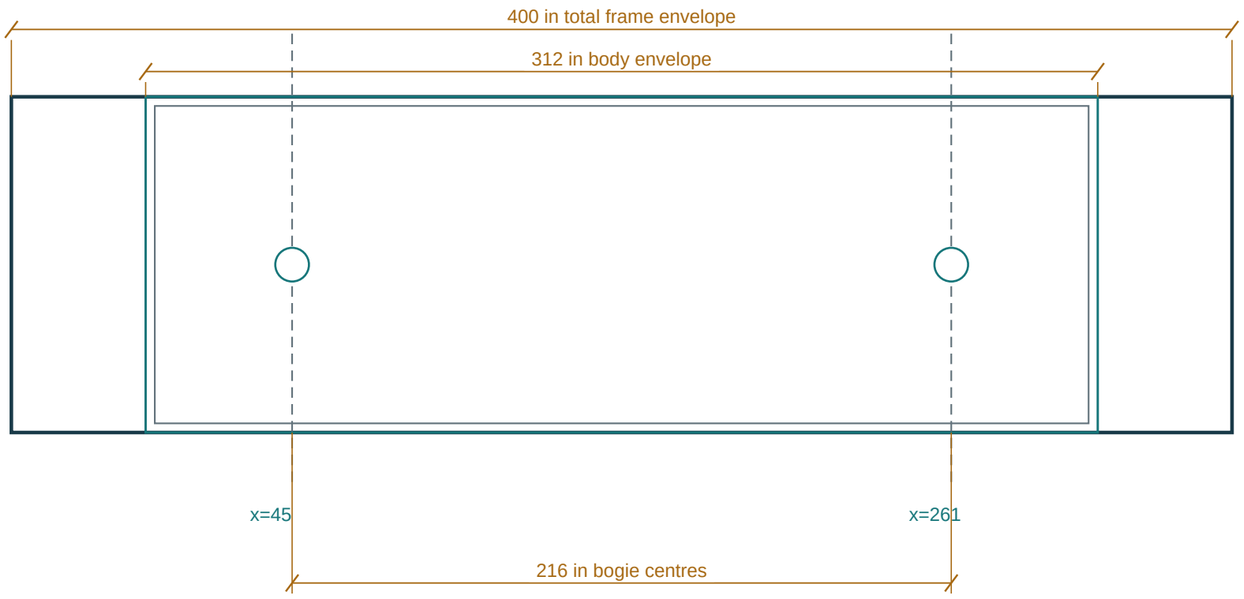
Prototype control geometry and common load cases

Phase 3B controls the inputs used by frame, bogie, brakes and couplings. Proposed upper cases are analysis inputs, not rated vehicle masses or permission to add tare.

G1 / REGULAR OR ACCESSIBLE COACH CONTROL DATUMS



x=0: front inside end wall. y=0: coach centreline. z=0: top of rail.

Floor z=28 in remains provisional; floor-local mockup datum is z=0 on M01/A01.

G2 / COMMON MASS CASES / CALCULATED

Regular coach upper case: 13,200 kg each. Accessible coach upper case: 11,000 kg. Locomotive + hotel allocation: 14,500 kg.

3 x 13,200 + 11,000 + 14,500 = 65,100 kg proposed upper train case.

Nominal / stress budgets remain 54,755 / 59,840 kg. These are not weighed masses, certified ratings or final allowable gross loads.

G3 / BASIS TO HOLD THROUGH PROTOTYPE

36 in gauge under evaluation. Owner target: 0% longitudinal grade; engineer sets survey acceptance and allowable deviation. Drainage remains independent.

The North Pole location is deferred. Maintain unloading, accessible boarding and rescue interfaces at both stations. Ride programming is outside this package.

G4 / MEASUREMENT CONTROL

Record frame diagonals and level datum before body installation. Survey bogie pivots, rail gauge and floor height in unloaded and loaded states. Compare measured mass / axle reactions to accepted engineering cases. Do not infer acceptance tolerances from this sheet.

G5 / DIMENSION RELEASE

Keep retained 44 in end platforms, 104 in clear width and 306 in clear length until mockup / equipment interface review. A deeper real frame or selected lift may require an explicit geometry revision. Do not silently carry study depths into shop drawings.

G6 / DOCUMENT PRECEDENCE

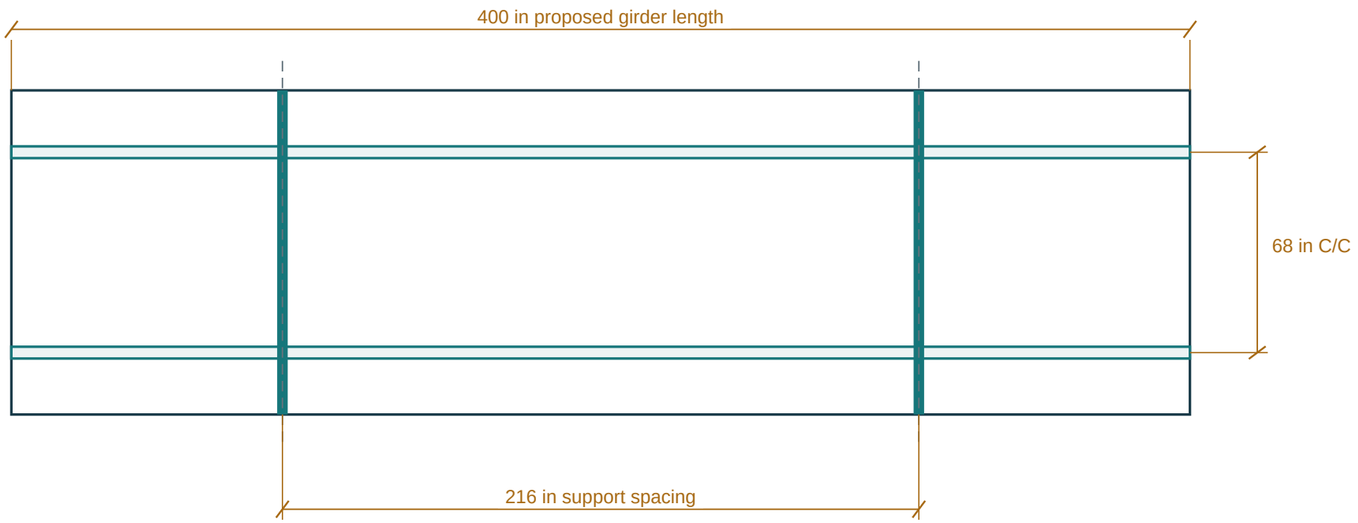
Fabricate passenger hardware only from responsible engineer / OEM approved drawings. G01 defines coordination inputs; F01-B01 are component development studies. M01/A01 are non-passenger mockup work instructions, not vehicle construction releases.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Frame candidates: girder layout, sections and height stack

Dimensioned candidate geometry for analysis only. No girder, crossmember, bolster, weld, fastener or material procurement is released by this sheet.

F1 / TWO-GIRDER CANDIDATE PLAN / INCHES

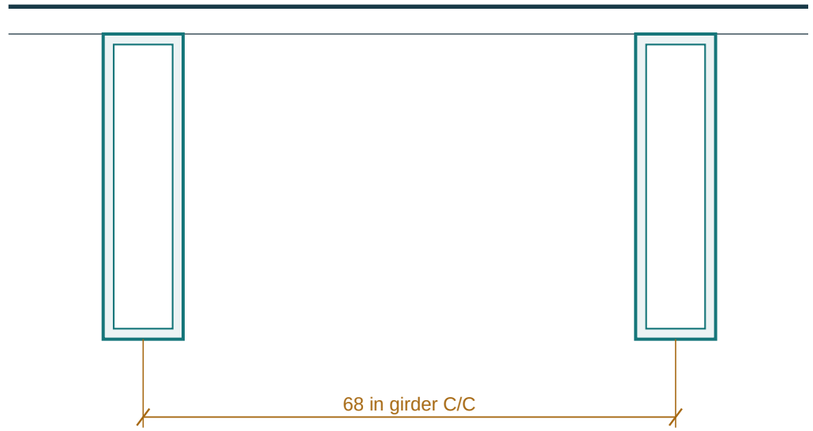


Girder centre lines y=-34,+34; nominal horizontal width4 in. All are study coordinates.

Crossbolsters x45 /261: section, connection and brake / coupling load paths HELD.

F2 / STRONG-AXIS SECTION / FLOOR STACK

Finished floor z28 study; 0.75 in panel allocation study only.



Top of girder z27.25. Underside = z27.25 - candidate depth.

8 /10 /12 in depths -> z19.25 /17.25 /15.25 in.

F3 / DOCUMENTED SECTION PROPERTIES - CANDIDATES, NOT A SELECTION

HSS nominal (in)	Design wall (in)	Weight (lb/ft)	Ix (in^4)	Sx (in^3)	Two 400 in girders (lb)
8 x4 x1/4	0.233	19.02	42.5	10.6	1,268.0
10 x4 x1/4	0.233	22.42	74.7	14.9	1,494.7
12 x4 x1/4	0.233	25.82	119	19.8	1,721.3

Atlas Tube rectangular HSS properties; actual grade, availability and design method require structural engineer confirmation.

STATIC CASE SCREEN / NO PASS LIMIT

14-case max stress for 8/10/12 in depths:220.3 /156.7 /117.9 MPa; max deflection 50.6 /28.8 /18.1 mm. Different cases govern. No dynamic factor, fatigue check or acceptance criterion is assigned.

FLOOR / BOGIE HOLD

Separate actual bogie mass from frame loads. Reconcile member depth with bogie clearance / floor height. Check local loads, torsion, fatigue, holes, attachments and jacking; beam screening cannot release the frame.

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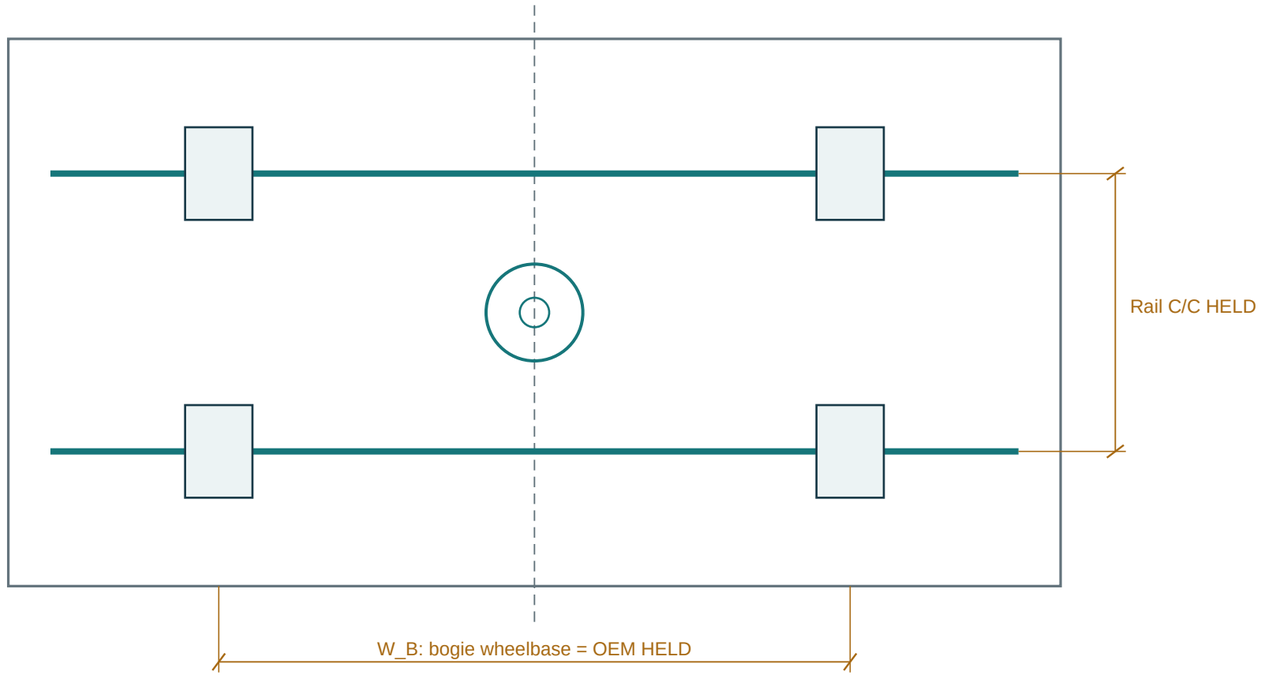
Bogie interface return drawing and measurement schedule

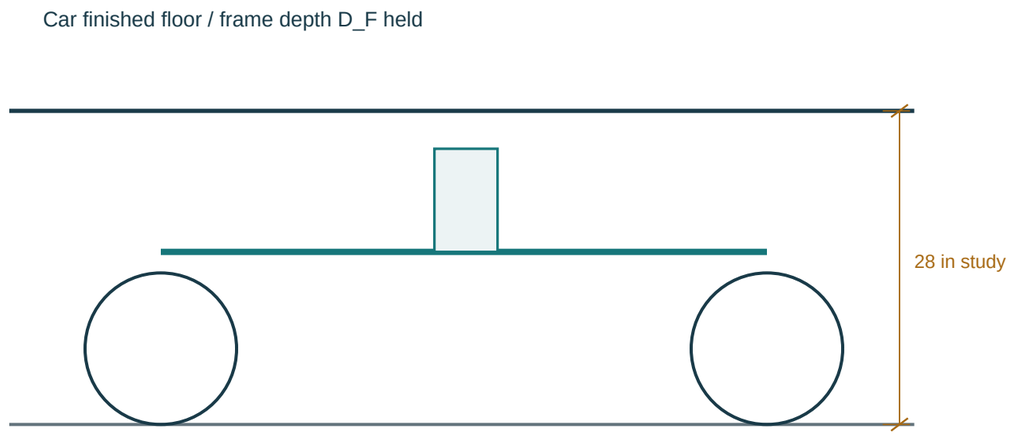
Preferred development path: obtain an engineered railway bogie / wheelset package. This return sheet requests certified geometry and ratings; no catalog match is claimed.

BG1 / PLAN - 36 in RAIL GAUGE BASIS



Support pivot / side-bearer location, rotation stop and brake envelope: OEM HELD.

BG2 / SIDE INTERFACE AND HEIGHT STACK



H_P pivot height, wheel radius r_W, jounce / rebound and side bearings HELD.

RETURN A / GEOMETRY + LOADS

A01: gauge and gauge-measurement plane. A02: wheel profile / back-to-back / flange limits. A03: wheelbase and all rotating envelopes. A04: pivot / side-bearer locations and loaded height range. A05: gross bogie rating and unequal axle / side load limits.

RETURN B / EVIDENCE

B01: bearings, axles and wheel material / inspection documentation. B02: spring characteristics / travel stops. B03: minimum curve and reverse-curve limits. B04: brake mounting and torque reaction limits. B05: drawings, service intervals, wear limits and traceability.

ACCEPTANCE GATE

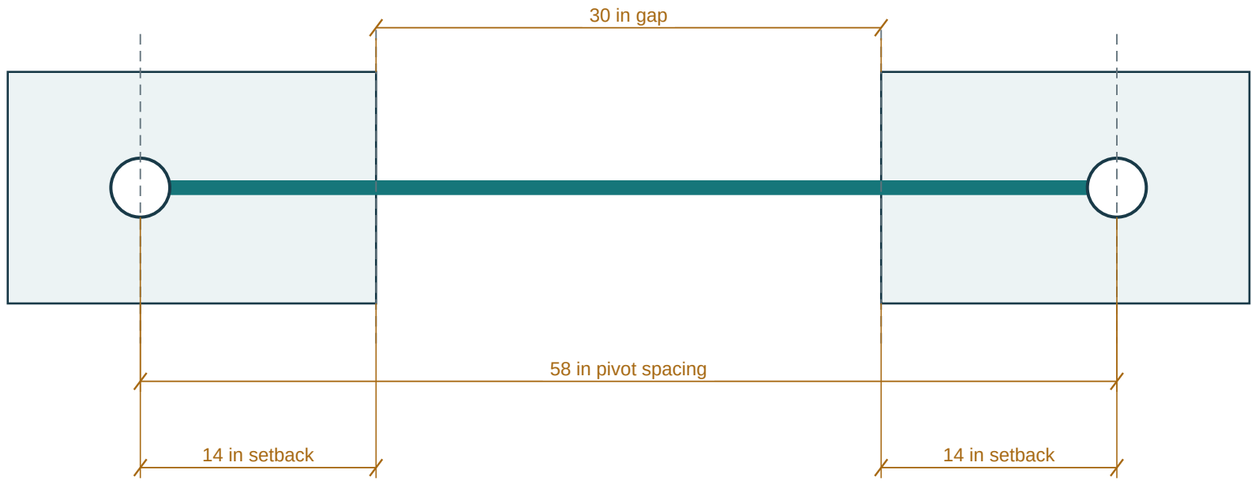
Return supplier-certified dimensional drawing and rating basis. Check body sweep, braking, attachment loads and maintenance removal. A 36 in gauge match alone cannot approve this interface. No drilling or bolster fabrication from symbols shown here.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Connection prototype: geometry fixture and supplier interfaces

Engineering-owned prototype only. Retained 30 / 14 / 58 in values are kinematic test inputs; pin, bushing, socket and fastener dimensions require documented component selection.

C1 / PLAN GEOMETRY FIXTURE - RETAINED SENSITIVITY



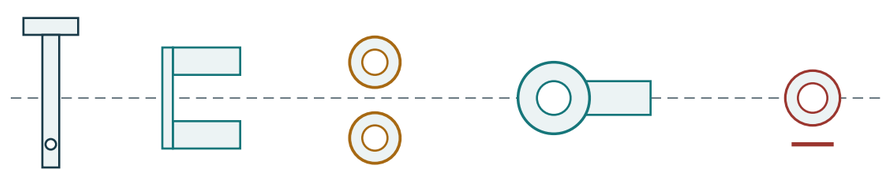
CTP02: record yaw / pitch / roll, stroke, hose sweep and platform pinch envelope.

C2 / RATED COMPONENT RETURN FIELDS

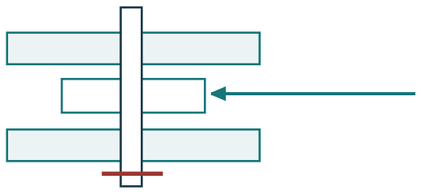
INTERFACES C-A THROUGH C-F

C-A: frame adapter hole pattern, load directions and attachment details. C-B: pivot-axis position and full articulation cone. C-C: drawbar rated draft / buff / fatigue spectrum. C-D: pin and bush material, fits, lubrication and wear limits. C-E: retention system and inspection access. C-F: emergency separation / services release provisions.

C3 / EXPLODED FUNCTIONS - NOT A MACHINING DETAIL



C4 / ATTACHMENT SECTION AND LOAD PATH



Pin shear / eye bearing / socket load spread HELD

TEST CTP01 / LOAD MEASUREMENT

Use reviewed load cell and fixture, calibrated and protected. Test draft, buff, slack and reversing only to engineer-issued levels and energy limits. No improvised pull test with occupied cars.

TEST CTP02 / MOVEMENT

Record mechanical stops, joint stroke, pin retention, alignment, full-service loops and guards. Combine vertical and horizontal articulation; independent yaw-only motion does not close the interface.

RELEASE HOLD

Dellner / Voith rail products demonstrate real coupling architectures, not a matched selection. Obtain supplier application acceptance and attachment loads before machining pins, lugs or frame sockets.

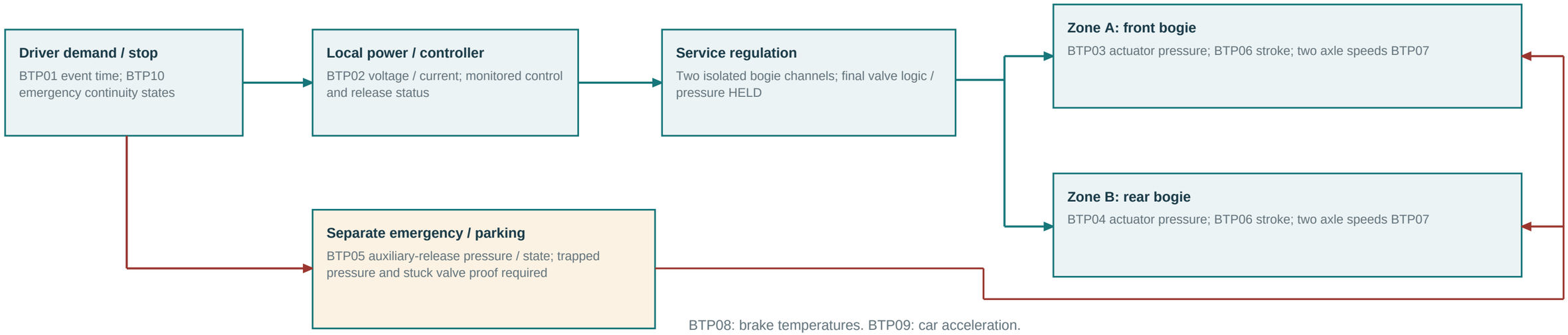
VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

PULSIPHER FARMS / CHRISTMAS RAILWAY

Brake prototype: two bogie zones and instrumented test points

Hydraulic service remains preferred. OEM must validate the selected service actuator and separate emergency / parking function; a passive SAHR candidate is not assumed suitable for modulated service.

B1 / ONE COACH - FUNCTIONAL TEST ARCHITECTURE



BTP08: brake temperatures. BTP09: car acceleration.

BTP07 includes independent ground speed to detect wheel slide.

B2 / CALCULATED FORCE SCREEN

Four braked axles per coach is an explicit study assumption. Regular 13,200 kg: service 6.60 kN / emergency 10.56 kN total at 0.5 /0.8 m/s2. Per axle 1.650 /2.640 kN if equally allocated. Accessible 11,000 kg: 5.50 /8.80 kN total. This is not a wheel-slide-safe distribution.

B3 / TORQUE + COMPONENT GATE

T_axle = F_axle x selected wheel radius. At illustrative r=0.25 m, regular emergency 660 N-m/axle; accessible 550 N-m/axle. Radius is not selected.

HSC1P10N rail spring-actuated candidate: manufacturer 20 kN force/disc,15 kg, max 45 mm disc width, two hydraulic release circuits. OEM force definition / service duty / fit remain HELD.

B4 / TEST ORDER + PASS CRITERIA

Bench: leak / pressure / stroke and powered / unpowered states. Roller or equivalent fixture: application timing, release and repeated thermal duty. Vehicle: reviewed unloaded then loaded stop tests on surveyed level track.

Engineer sets instrument accuracy, sample rate, sequence, fault injections, exclusion zone, runout and numerical acceptance.

SAFETY FUNCTION HOLD

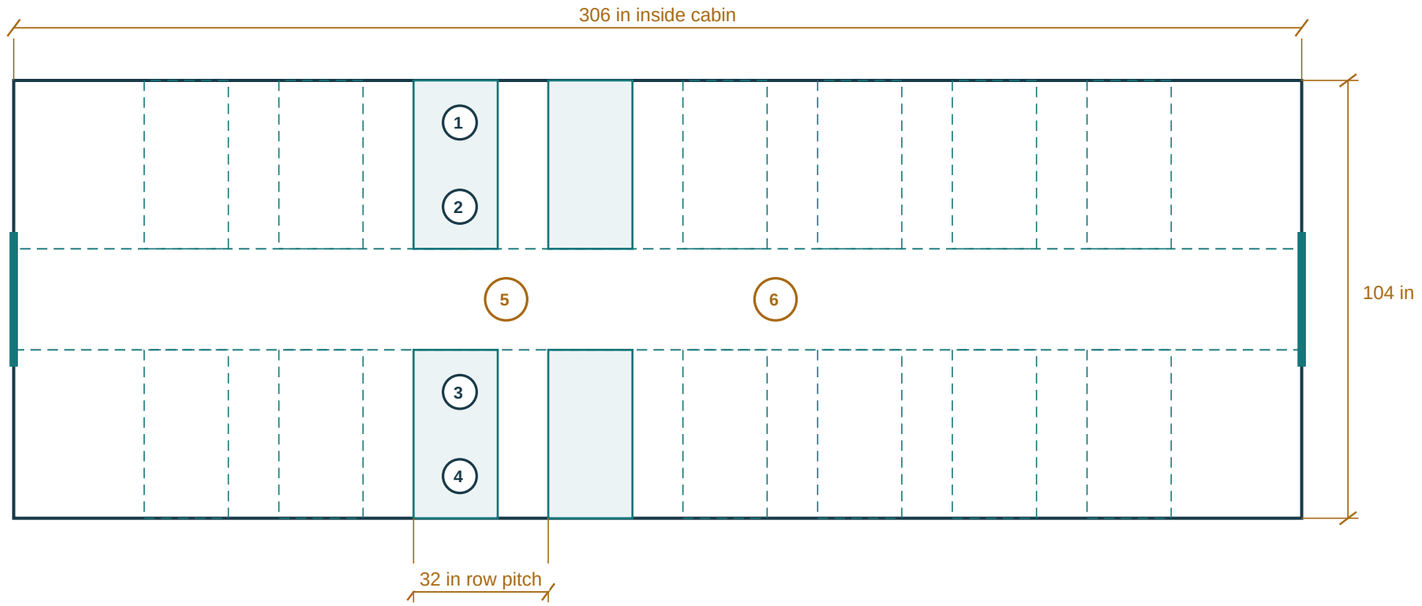
The HSC1P10N is passive spring-applied / hydraulically released. Obtain OEM acceptance for full-duty modulated service, or use an active service actuator plus independent SAHR package. Do not connect this drawing as a final hydraulic schematic. No automatic restart; controlled manual release must lock out traction.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Regular coach: six-adult full-scale mockup work order

Mockup only on stable level shop floor. Tape outlines first; any raised seats / temporary supports must be independently safe for the intended people and loads.

M1 / TAPE AND MOCKUP PLAN - ALL DIMENSIONS IN INCHES



Solid fill: build mockup rows x95 and x127; dashed boxes: tape all other benches.

Bench envelopes: 40 wide x20 deep; mockup seat top18 above local floor. Occupants face +x.

M2 / PERSON POSITIONS

Seated 1-4: centre x106; y=-42, -22, +22, +42. Standing 5: x117,y0. Standing 6: x181,y0. These are test positions, not validated human envelopes.

Measure real participants including coat / footwear and reach. Move through the car after the static trial; do not treat circles as clearance proof.

M3 / CRITICAL CHECKS

Row x95 seat occupies95..115. Trial knee extension115..124.5 leaves only 2.5 in before row 127; this is a study screen, not a pass.

Measure seat comfort, knee clearance, aisle passing, staff access, door operation and emergency exit reach. Record conflicts and proposed layout change; retain six-adult photo evidence.

M4 / BUILD SEQUENCE

1. Establish origin and full 306 x104 rectangle. 2. Tape aisle, 16 benches and end doors. 3. Add stable mockup benches at x95 and127. 4. Check seat top height / edges. 5. Conduct six-adult trial with observer and clear stop instruction.

M5 / RECORD SHEET

Record participant dimensions, measured knee gap, usable aisle width, ingress time observations, obstructions and comfort comments. Photograph seated and standing positions. No passenger rating or egress approval results from a mockup alone.

M6 / HANDOFF

Provide marked-up plan and measurements to body / accessibility engineer before freezing seating or ordering doors. Retain 32 guests as a planning assumption until circulation, occupant load and emergency exit requirements are resolved.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

PRELIMINARY ENGINEERING DEFINITION - NOT FOR FABRICATION OR CONSTRUCTION

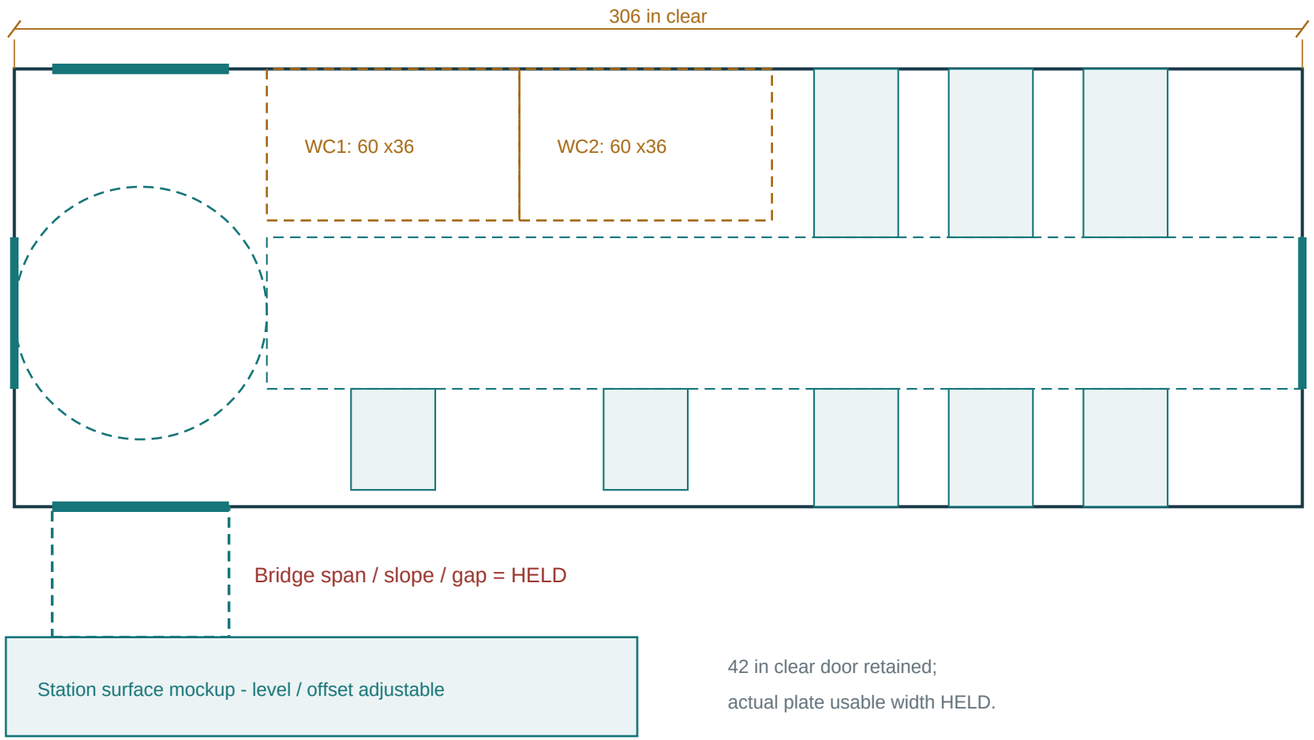
PF-TRAIN-S3B-R0 | 2026-10-17 | Prototype development studies. No engineer approval or manufacturing release implied.

PULSIPHER FARMS / CHRISTMAS RAILWAY

Accessible boarding mockup: bridge-first and lift conflict

Retain Stage 2 circulation geometry for testing. Normal boarding is required at both stations. Independent stalled-train evacuation remains a separate release gate.

A1 / RETAINED FLOOR PLAN AND BRIDGE INTERFACE MOCKUP



42 in clear door retained;
actual plate usable width HELD.

A2 / BRIDGE TRIAL MEASUREMENTS

Use occupied mobility devices and companions. Measure clear opening with hardware, gap / step, plate seating / restraint, route width, turning, threshold forces and emergency reversal.

Do not transfer a wheelchair onto an unsupported prototype bridge. Use a rated supervised test fixture or an engineer-approved supported mockup.

A3 / OPTIONAL LIFT - CONFLICT TO RESOLVE

48 in opening is an unintegrated alternate, not a revision to retained CAD.

A4 / MOCKUP HOLD POINTS

Mark 60 in turning circle centre(30,6); route y=-12..24; doors x9..51. Trial entry, turn, park, exit and rescue access. Record actual occupied device dimensions and accessory protrusions.

A5 / LIFT EVIDENCE

Braun reference requires 45.85 in minimum clear width and 50.96 in overall width. A 42 in retained opening does not fit. Even a 48 in opening needs frame, sidewall, door and maneuvering redesign; do not infer fit from width alone.

A6 / RELEASE / BB REGISTER

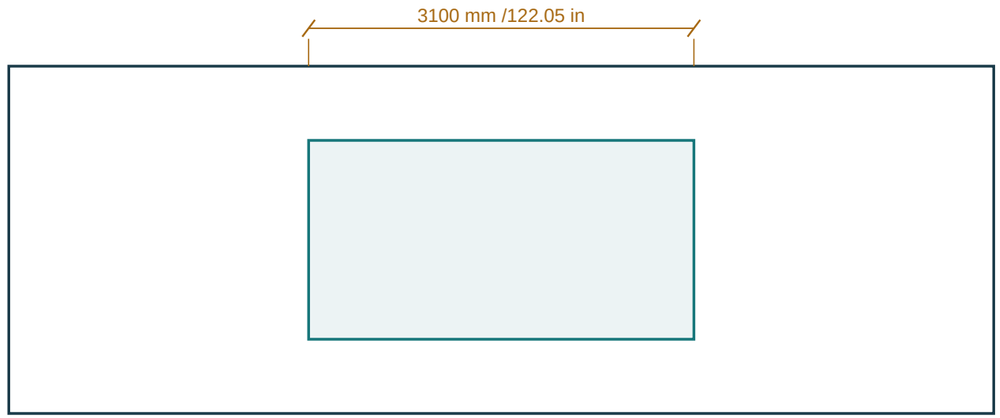
Confirm device ratings, anchors, interlocks, boarding slope / gaps, accessibility basis and independent rescue. Vehicle lift stow volume / motion and power-loss operation require OEM evidence.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

Utility packages: dimensional reservations and fit conflicts

Manufacturer envelopes are documented comparison candidates. Roof loads, thermal duty, service access, electrical supply, mounts and actual delivered mass require coordinated acceptance.

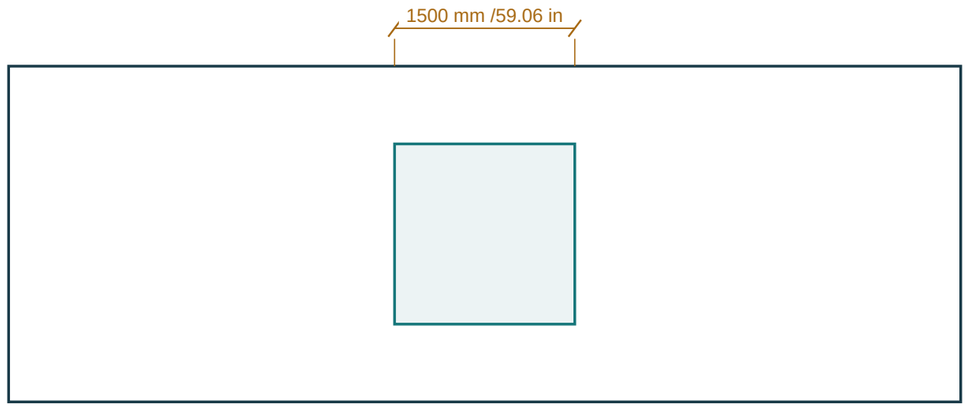
U1 / REGULAR COACH ROOF - KONVEKTA 7403 CANDIDATE



STUDY centre x153,y0; remaining 23.50 in each side is geometric space,

not a maintenance clearance. Manufacturer service envelope HELD.

U2 / ACCESSIBLE COACH ROOF - 5401 CANDIDATE



5401 envelope:1500 x1500 x370 mm;275 kg reference mass.

Capacity at listed catalog condition does not prove Nevada design duty.

U3 / HEIGHT AND MASS RECONCILIATION

Retained roof reference z116.614 in. 7403 adds16.339 in,

giving z132.953 in plus unknown curb / isolators. Overall height HELD.

U4 / CONFLICTS BEFORE PURCHASE

Regular 7403 mass 520 kg leaves only 30 kg in the entire 550 kg HVAC allocation for curb, mounts, ducts, wiring and drains; do not accept as a demonstrated mass fit.

A 15 in interior air handler under the retained 86.61 in ceiling leaves 71.61 in headroom. Reject that beneath-aisle location; other locations need redesign.

Any underfloor condenser needs complete bogie / brake / stair / track clearance and heat-rejection review.

UTILITY CHECK UTP01-03

UTP01: source transfer isolation / no backfeed. UTP02: protective-earth continuity through each vehicle and connector. UTP03: local isolation / branch protection and fault-current design. Qualified electrical lead defines methods and limits.

UTILITY CHECK UTP04-06

UTP04: loaded voltage / current and startup transients. UTP05: cooling / heating, ventilation and door-cycle temperatures. UTP06: emergency-light autonomy with normal supply removed. Do not infer battery size from HVAC power.

PACKAGE ACCEPTANCE

Confirm actual service openings, drain routes, removable panels and lifting access using supplier envelope drawings. Freeze roof reinforcement only after mass, location and mount load cases are accepted. Refer PU register.

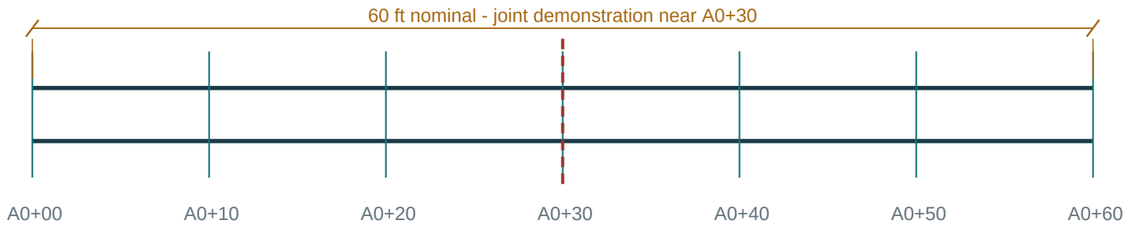
VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release

PULSIPHER FARMS / CHRISTMAS RAILWAY

Track pilot work order: two disconnected training modules

Training / workmanship trials only. No train operating authority. 0% longitudinal grade is the target; survey acceptance, track system and tolerances require the track engineer.

T1 / MODULE A - NOMINAL 60 ft TANGENT



Final cuts, thermal gaps, tie positions and joint support detail HELD.

Rail lines symbolic: measure actual gauge at the accepted railhead plane.

T2 / MODULE B - NOMINAL 24 ft REFERENCE ARC

Study radius 50 m /164.042 ft. Not a permanent route or approved radius.

No transition fabricated. Keep Module B physically disconnected from A.

10 ft chord mid-ordinate0.9146 in: teaching geometry, NOT an acceptance tolerance.

T3 / CREW SEQUENCE AND WITNESS POINTS

H1	**Before materials ordered**	Accept rail / tie / fastening system, tools and method.
H2	**Before formation covered**	Verify levels, bearing and drainage; record survey.
H3	**Before rail fastening**	Accept unloaded bend / joint first article and geometry.
H4	**Surfacing / as-built acceptance**	Check every tie; record line, level, gauge and crosslevel.
H5	**Before repeat crew work**	Independent pilot acceptance and competency review.
H6	**Before loading or movement**	Separate approved restraint, runout and recovery plan.

T4 / MATERIAL PLANNING ONLY

84 nominal route ft /168 nominal rail ft. Six 30 ft rail blanks =180 rail ft planning allowance: four for tangent A, two for curve B. Final end crop, bending loss, joint placement and purchase quantity are HELD.

Select rail / tie / fastening system before procuring or cutting. No consumer laser or tape-only check substitutes for engineer-defined track acceptance.

T5 / STATION INTERFACE WITHOUT A SITE

Keep a separate boarding-gap / height mockup at the shop. Confirm accessible bridge interface and stopping reference before permanent station edges. North Pole location stays deferred.

Rolling tests require a separate reviewed restraint, exclusion, runout and recovery plan. These modules alone cannot qualify the passenger train.

VERIFIED = source data | CALCULATED = stated-input screen | CANDIDATE / STUDY = proposed for review | HELD = required before release